

Homework (Habanero)**Q1.** Convert $x^2 + 6x + 8$ to vertex form.

- (A) $(x + 3)^2 - 1$
 (B) $(x + 3)^2 + 1$
 (C) $(x + 4)^2 - 8$
 (D) $(x + 4)^2 + 8$
 (E) $(x + 3)^2 - 9$

Q2. What is the vertex of the parabola $x^2 + 4x + 5$ after completing the square?

- (A) $(-2, 1)$
 (B) $(-2, -1)$
 (C) $(-2, 3)$
 (D) $(-2, 4)$
 (E) $(-3, 2)$

Q3. The dosage of a new medicine can be modeled by $d(t) = t^2 + 2t + 1$, where d is the dosage and t is the time in hours. Convert this function to vertex form.

- (A) $d(t) = (t + 1)^2$
 (B) $d(t) = (t + 1)^2 - 1$
 (C) $d(t) = (t + 1)^2 + 1$
 (D) $d(t) = (t - 1)^2$
 (E) $d(t) = (t - 1)^2 - 1$

Q4. The growth rate of a cell culture is given by $g(x) = x^2 - 4x + 3$, where g is the growth rate and x is the time in days. Find the vertex form of this function.

- (A) $g(x) = (x - 2)^2 - 1$
 (B) $g(x) = (x - 2)^2 + 1$
 (C) $g(x) = (x + 2)^2 - 1$
 (D) $g(x) = (x - 3)^2 - 4$
 (E) $g(x) = (x + 3)^2 - 4$

Q5. A patient's temperature after taking a dose of medicine can be modeled by $t(x) = x^2 - 2x - 3$, where t is the temperature and x is the time in hours. Complete the square for

this function.

- (A) $t(x) = (x - 1)^2 - 4$
 (B) $t(x) = (x - 1)^2 + 4$
 (C) $t(x) = (x + 1)^2 - 4$
 (D) $t(x) = (x + 1)^2 + 4$
 (E) $t(x) = (x - 2)^2 - 5$

Q6. A biologist is studying the population growth of a new species, which can be modeled by $p(x) = x^2 + 6x + 5$, where p is the population and x is the time in years. Convert this function to vertex form.

- (A) $p(x) = (x + 3)^2 - 4$
 (B) $p(x) = (x + 3)^2 + 4$
 (C) $p(x) = (x - 3)^2 - 4$
 (D) $p(x) = (x - 3)^2 + 4$
 (E) $p(x) = (x + 4)^2 - 11$

Q7. The spread of a new disease can be modeled by $d(x) = x^2 - 8x + 12$, where d is the number of people infected and x is the time in weeks. Complete the square for this function.

- (A) $d(x) = (x - 4)^2 - 4$
 (B) $d(x) = (x - 4)^2 + 4$
 (C) $d(x) = (x + 4)^2 - 4$
 (D) $d(x) = (x + 4)^2 + 4$
 (E) $d(x) = (x - 5)^2 - 13$

Q8. The dosage of a vaccine can be modeled by $v(x) = x^2 + 2x - 6$, where v is the dosage and x is the time in months. Convert this function to vertex form.

- (A) $v(x) = (x + 1)^2 - 7$
 (B) $v(x) = (x + 1)^2 + 7$
 (C) $v(x) = (x - 1)^2 - 7$
 (D) $v(x) = (x - 1)^2 + 7$
 (E) $v(x) = (x + 2)^2 - 10$

Open Ended Questions (FRQ)**Q9.** A new medicine's efficacy can be modeled by $e(x) = x^2 - 4x + 3$, where e is the efficacy and x is the dosage. Complete the square and interpret the transformed equation.**Q10.** The growth rate of a cell culture can be given by $g(x) = x^2 + 2x + 1$, where g is the growth rate and x is the time in days. Convert this function to vertex form and explain the significance of the vertex in this context.**Chef's Tips**

- Ensure students understand the concept of completing the square and its application in real-world scenarios.
- Encourage students to interpret the transformed equations in the context of the given problem.
- Remind students to check their work and ensure accuracy in their calculations.